

How Do Economists Measure Inequality, and what do these measures miss?

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1 Introduction

Economic inequality refers to the difference in the distribution of wealth or income within a group of people [1]. Usually, we measure this in terms of income inequality, namely with the use of the Lorenz curve which plots the cumulative share of income against the cumulative share of the population. We also frequently use the Gini Coefficient, especially to compare inequality between economies, which is calculated using the Lorenz curve $G = 1 - 2 \int_0^1 L(p) dp$ [2] [1]. The Lorenz curve plots the cumulative share of the population against the cumulative share of income or wealth. A perfectly equal society would have a straight diagonal line as every marginal household has the same marginal income, so the gradient will be the same. The Gini Coefficient quantifies this inequality effectively, ranging from 0 (perfect equality) to 1 (perfect inequality), with the UK at 0.36 [1] [2].

In addition to income inequality, economists also measure wealth inequality (distribution of assets) and inequality of opportunity. Inequality of opportunity can measure how much a child's eventual income is determined by drivers beyond their control (family background, ethnicity, etc.) [3]. These measures of inequality are usually positively correlated [33], but overall, they “offer different yet complementary insights” [3] into how unequal a society is [3].

Simple income statistics often miss non-monetary dimensions of living standards [5], so other composite measures can measure more subjective drivers, including wellbeing. For example, the Human Development Index (HDI), created by the United Nations, combines drivers including health (life expectancy), education (years of schooling), etc. to give an overall description of development [4]. The HDI is designed to go beyond GDP [4] “to emphasize that people and their capabilities should be the ultimate criteria for assessing development, not economic growth alone” [4]. Finally, quality of life (QoL) is an even broader, yet more subjective measure. It takes some objective drivers (material resources, health, living conditions, environment, etc.), but also includes subjective drivers of wellbeing [5].

Historically, inequality has been measured in many different ways, albeit most were ineffective. William Petty was an early economist in the 17th century, who tried to quantify population, wealth, and tax, and made an early attempt to compare social groups numerically [23]. After that, in 1688, Gregory King tried to make a social table to formalize occupational groups and incomes, in order to infer inequalities between classes [24]. Earlier censuses also provided raw material, including the Renaissance fiscal catasti (land registries (translated from Latin)), specifically the Florentine catasto of 1427, as well as hearth-tax rolls from even earlier, which gives historians the necessary information for describing inequality in pre-industrial societies [6] [7]. In the nineteenth century, social statistics became a more advanced field, specifically with Adolphe Quetelet's

“social physics”, and more commonplace collection of statistics by the government, which allowed much easier and more accurate comparisons and the creation of early indices. [25] [29] [28].

2 Income-based Measures of Inequality

Economists most commonly measure inequality using income distribution statistics. The Gini Coefficient in combination with the Lorenz curve are commonly used tools [28], with the Lorenz curve being more than a century old (1905) [26]. As described in the Introduction, the Lorenz curve shows the cumulative share of income (y-axis) held by the cumulative share of the population (bottom share of the population, x-axis). In a perfectly equal society this curve would be the 45° “line of equality” [2] (each percentile of the population would hold the same percentile of income) [2]. In practice, Lorenz curves are always below the line, with the area between them summarizing inequality. Economists use this single number to compare inequality. Corrado Gini [27] originally defined this measure in 1912 and today the Gini index is widely used for international comparisons [40] [8].

Another dimension is how income inequality persists across generations. This is measured by Intergenerational Income Elasticity (IGE) [42]. Defined by regressing $\ln(y_{child})$ on $\ln(y_{parent})$; the slope is the IGE. The rank–rank slope uses percentile ranks and is more robust to outliers. An IGE of 0 indicates no persistence (parental income does not predict children’s income, full mobility). An IGE of 1, by contrast, means full persistence (children’s incomes move one-for-one with parents’ incomes) [43]. For example, early US studies found that the IGE was approximately 0.4 [9]. In practical terms, a high IGE (say 0.5) means that if parents’ income is 10% higher than the average, children’s income is also 5% higher than the average. This demonstrates inequality of opportunity. IGE specifically is calculated using a logarithmic regression correlation between incomes across generations, so it isn’t as simple as this, but the principle is the same. Such mobility metrics complement other measures by demonstrating how elastic the economy is.

Other indices exist too, such as the Theil index and Atkinson index, which also quantify inequality with different sensitivities to tails of the distribution [34]. Theil is based on information entropy [30], it specifically talks about the “surprisal” [68] [36] of observed incomes relative to perfect equality [30]. It is measured using the formula [35]:

$$GE_{\alpha} = \frac{1}{\alpha(\alpha - 1)} \left(\frac{1}{n} \sum_{i=1}^n \left(\frac{y_i}{\bar{y}} \right)^{\alpha} - 1 \right), \quad \alpha \neq 0, 1.$$

In the limit $\alpha \rightarrow 1$, this becomes the Theil index.

It’s useful as it can split easily into within-group and between-group components, making it ideal for analysis over larger systems [28]. It is also beneficial because it is a formal analogy to Boltzmann/Gibbs entropy in statistical mechanics, and as such, can much more easily demonstrate progress over time, similar to entropy in the field of physics. In this case, inequality is a deviation from maximal disorder (perfect equality) [32] [30] [31] [34] [35] [36] [37]. Meanwhile Atkinson (1970) introduced a family of indices sensitive to poverty aversion. However, the Gini has remained dominant because many national statistics offices report it, and it is relatively easy to interpret [44].

2.1 Evaluation

One shortcoming is that the Gini Coefficient and the Lorenz curve only capture distribution of income, they ignore drivers like absolute income levels, health, or leisure that also affect wellbeing and happiness. This limits their ability to aid in international comparisons. For example, having a high income does not guarantee good health, education, or environmental quality. By contrast, households with low incomes in countries with universal health care or education can still enjoy a relatively good life. Quality of life is “a broad concept” encompassing both objective drivers (e.g. health and living conditions) and subjective perceptions (happiness) [5]. A simple comparison between incomes is not capable of measuring such drivers [5].

Even when it comes to just comparing incomes, Gini and similar statistics have drawbacks. The Gini Coefficient, being a single number, does not demonstrate some crucial differences [38] [34] [28]. Two countries with different income distributions can have the same Gini Coefficient. For example, an economy where income inequality stems from low incomes in the bottom 50% could have the same Gini Coefficient as an economy where 50% of the income goes to only a small share of the population, as shown with the comparison between Diagram 1 and Diagram 2 [45] [28].

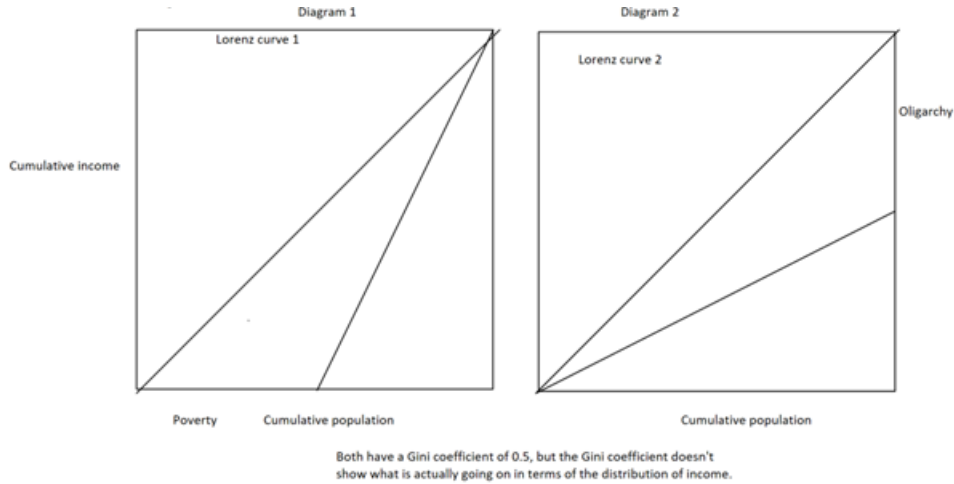


Figure 1: Lorenz curve and equality line.

In addition to this, the Gini Coefficient does not indicate absolute living standards. A developed country and a developing country can have identical Gini Coefficients, even when, by comparison, one has much better quality of life than the other. Some economic studies, such as Hauser et al. (2022) [10], show that alternative multi-parameter measures such as the Ortega index [10], capture such nuances more effectively than even the Theil and Atkinson indices, especially compared to the Gini Coefficient [10]. Overall, purely income measures miss drivers such as health, education, social mobility, and freedoms that affect the actual wellbeing within the population [46] [47] [66].

3 Non-income measures

The United Nations and related agencies provide much broader inequality measures which do make use of such drivers. The most notable such index is the Human Development Index, which

includes aforementioned drivers such as health (life expectancy at birth), education (mean and expected years of schooling), and standard of living (GNI per capita) [4]. The HDI is written in a similar manner to the Gini Coefficient, being a number between 0 and 1, albeit in the opposite direction, with 1 being the best HDI, while 0 being the worst. This composite figure demonstrates that a country with medium income, but high education or health care can rank similarly to a wealthier country with poorer social services.

In addition to this, the United Nations Development Programme (UNDP) introduced an inequality-adjusted HDI (IHDI) in order to further take into account inequality. The IHDI takes the geometric mean of health, education and income indices, then “discounts” them by the level of inequality in each dimension [11] [48]. This means that the $IHDI = HDI$ under perfect equality, but will otherwise always be below HDI, even more so when inequality is high [11]. For example, if Country A and Country B have the same average health and education drivers, but A has a large health inequality, A’s IHDI will be lower. As such the IHDI penalizes countries with unequal distributions of quality of life, giving a much more complete demonstration of quality of life overall within the entire population.

Furthermore, the UN has adapted Sustainable Development Goals (SDGs) since 2015, with the background behind them originating in the 1970s. The UN uses “231 indicators” (248 including duplicates) to measure progress within them, many of these indicators including inequality in various areas, and different ways of measuring it. One of these goals, Goal 10, emphasizes the need to “Reduce inequality within and among countries” [12], and in its first target (10.1), calls to “progressively achieve and sustain income growth of the bottom 40 per cent of the population” [12], while other targets and goals address social protections and fiscal policy [12]. The official UN description also emphasizes reducing inequalities based not only on income, but also on age, gender, race, disability, and other status [12]. As well as this, many other UN Goals focus on “access to water and sanitation” as well as energy, meaning that inequality in measures not even covered in IHDI or Goal 10 are still covered by the UN’s targets and indicators.

Education and access measures are also beginning to be used. The UN Education Goal, Goal 4, uses metrics including expected years of schooling and literacy rates to calculate inequality in learning and education. Economists believe that such inequalities can be as, if not more, important than income inequality [46]. Although this is not necessarily inequality being measured, information such as enrollment ratios and test score gaps are also monitored to measure opportunity inequality between groups [49].

3.1 Evaluation

The UN tracks such goals using national statistics, however, such data is not necessarily trustworthy. UN agencies rely on member states to do their own surveys, censuses and data collection to compute indices like HDI or SDG progress. However, many countries have gaps in their data or may purposefully skew it [53]. UNDP has noted that “gaps in the availability of disaggregate socio-economic data” are common in countries worldwide [13]. Meanwhile, some other governments may lack the capacity or willingness to collect detailed statistics, leading to outdated or incomplete reporting. For instance, household surveys that feed into HDI education indices might only be collected once every decade in some developing countries [13]. As well as this, some statistics are underreported, such as income shares of the elite [13]. Overall, while

UN frameworks encourage broad measures of inequality using many separate indicators, the underlying data is usually weak or non-existent [13].

In contrast, some UN indices set useful standards [50]. This is supported by the fact that most countries compile HDI and report it annually and, as such, we have a common benchmark for development and inequality. The SDGs’ emphasis on inequality has also led to investments into more effective method of data collection (e.g. crowd-sourced health and education monitoring) [51]. Even if imperfect, these frameworks nudge governments to consider multiple dimensions contributing to inequality. Despite this, economists caution that the headline UN figures cannot always be trusted, especially when data collection is indirect [13].

4 What does the future hold?

Economists and international organisations have proposed new metrics to complement or even replace the Gini Coefficient and the HDI [47]. Some of these measures focus on opportunity-based indicators. For example, the Human Opportunity Index (HOI), which was developed by the World Bank and UNICEF, measures how basic opportunities (such as education, water, sanitation, etc.) are distributed relative to the overall income of a society [14]. The HOI is high when access is universal and lower when access depends on circumstances such as family, location, or income [14].

Private and Public opportunity indices (often on education, infrastructure, and social mobility) are being created for analysis within a larger economy (e.g. in the US) to aid policymakers at the state or city level when making decisions [52].

Beyond these, new composite indicators have been developed. Namely, the Social Progress Index (SPI), which is an independent index that evaluates countries based on 57 non-economic measures, including, but not limited to, basic needs (nutrition and shelter), wellbeing (health, education, environment) and opportunity [15]. The SPI explicitly goes “beyond economic metrics” to capture social and environmental progress [15].

Other proposals include variations of the Gini Coefficient that focus on wealth or consumption (rather than income), and “inequality-adjusted” versions of poverty indices [34]. In addition to this, another potential method to measure inequality in wellbeing between countries is through the use of Bhutan’s Gross National Happiness [54] [55], which is a well-known national effort to measure wellbeing in nine areas (psychological wellbeing, health, education, culture, etc.), demonstrating another multidimensional approach [55].

4.1 Evaluation

However, economists believe that a large volume of statistics can cause confusion and strain resources in governments [47]. Each new index requires reliable data, which is usually scarce in many countries (as mentioned above, even for existing indices, in some countries the data is unreliable and only available once or twice a decade). The UNDP also agrees that many countries already struggle to gather basic socio-economic data [13]. As such, creating more metrics would likely overwhelm statistical agencies. Too many overlapping indicators would also decrease the ability to understand inequality and would likely lead to corruption, with governments choosing which indicator hides the most inequality and reporting that one [56]. This would compound with the other issue of data collection becoming inconsistent, as countries will now have different

measures which are harder to compare, as they are different in some respects. For example, if IHDI, SPI, Gini Coefficient and GEI are tracked simultaneously, it will be hard for governments to choose which to prioritize and use as the base unit of measurement. Furthermore, new composite indices mean that choices have to be made in the weights of components and the methods for collecting the data, and such choices could also contain normative statements that contain value judgments, even if this is not obvious. As Hauser et al. (2022) note, any single parameter measure (such as Gini) necessarily loses information, and creating more indices does not necessarily solve that problem [10]. As such, a balance is necessary, broad measures should inform and be easy to collect, but raw data can also be used for comparison within a country in order to guide policymakers.

Disinformation and media effects can negatively affect the reliability of happiness-based indicators. Subjective wellbeing surveys, such as GNH or other measures of happiness, depend on people’s beliefs about their situation, as much as on facts if not more, and these beliefs can be manipulated [57]. Fake news can spread quickly online [58], potentially changing peoples’ opinions of their safety and happiness. This also means that people’s beliefs can either be inflated if they receive positive information, or negatively affected, if they receive negative information. As such, measured happiness is very unreliable and can potentially even be manipulated due to its subjective nature. The spread of fake news is accelerating [16], especially with the rise of AI-targeted content [59], while use of social media during crises has been linked to increases in anxiety and depression, which are the measures that happiness indicators take into account [17].

Furthermore, measuring National Happiness conflicts with the Easterlin paradox and socio-economic comparison effects. Having more money does not necessarily increase happiness [61]. Easterlin, an early researcher of happiness [60] showed that rising incomes over time do not necessarily raise life evaluation, especially once adaptation and changing aspirations are accounted for [62]. Kahneman and Deaton found that higher income raises life evaluation but not experienced wellbeing beyond a certain point [20]. This implies that money buys wellbeing to some degree, but not for all kinds of welfare [19].

On another note, people evaluate themselves relative to their peers, as implied by Festinger’s social-comparison theory [21]. As such, a rise in GDP can increase the reference level but leave measured happiness unchanged or even lower if inequality or relative status worsens [19]. Empirical studies confirm strong comparison-income effects, as subjective wellbeing changes in correlation with relative, not necessarily absolute, income according to study [22].

5 Conclusion

Overall, economists have developed many tools that measure inequality, but each tool has its limits. A single Gini Coefficient, or even a more detailed Lorenz curve only demonstrates the income distribution, and potentially hides other drivers affecting inequality [10]. Even broad composite indices such as the HDI “do not reflect on inequalities, poverty, human security, empowerment, etc.” [4]. In practice, a single number can’t capture all the drivers affecting inequality, but those problems also suggest that there could be a better approach. Namely, some multi-dimensional indices can combine many indicators, such as those for opportunity, health, education, and equality, including the newer IHDI. An example of this would be to combine the Gini Coefficient with measures of educational access (e.g. schooling years gaps) and health disparities (e.g. infant mortality gaps) to provide a fuller picture of equality within a society.

This could even be brought to Lorenz curves, which are currently mainly just used for income and wealth, but could also be used for other measures, such as health. Although a graphical tool, the Concentration Curve, already exists for showing health inequality, the Concentration Index is hard to calculate. As such, here is my attempt of a curve for the inequality of life expectancy in the UK, similar to the Lorenz Curve, and to the Concentration Curve. This curve uses statistics acquired from the ONS and demonstrated health inequality in the UK. There are no diagonal or other features of the Lorenz curve, as in some quartiles health increases at a greater rate compared to others, it isn't necessarily possible to show this as a cumulative measure in the same way as income. However, a unit measure can still be derived from this, similar to the Gini Coefficient, by taking the rectangle from (0, 0) to the highest point on the graph, and then dividing the area above the curve in that rectangle by the total area of the rectangle. The result is that, if we assume the range of life expectancy to be the same for 2 populations, those with a higher usual life expectancy will have a lower value, meaning that the value reflects on real happiness as well (a change in 10 years at a life expectancy of 100 is 10%, while at a life expectancy of 50 it's 20%) [63].

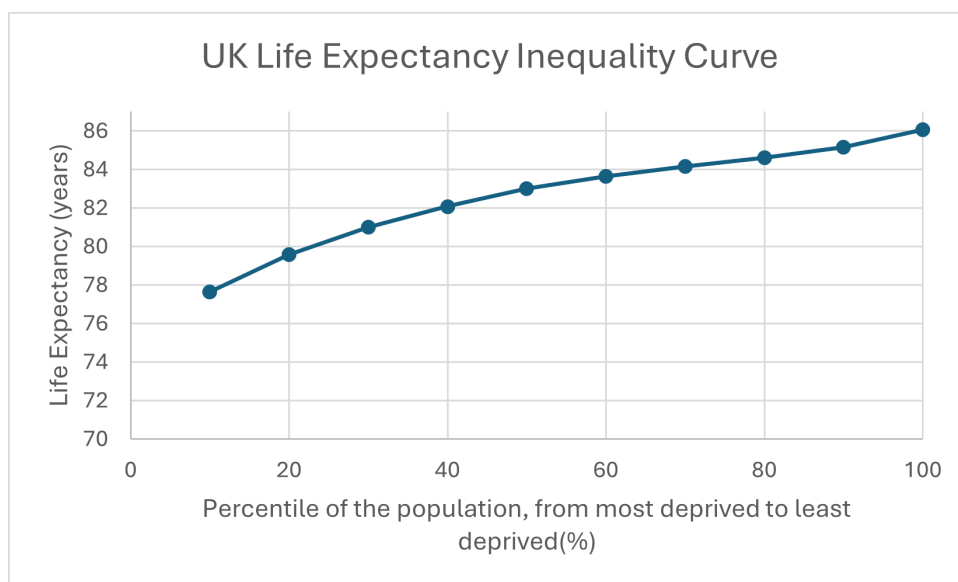


Figure 2: UK Life Expectancy Inequality Curve 1

Meanwhile, for inequality of opportunity, the rank–rank slope already exists and helps see which sections of the population have the most opportunities, and which are the most unequal compared to the rest of the population.

In summary, economists measure inequality mainly via the Gini coefficient and relative income indices, but these miss non-income drivers of welfare [5] [8]. The UN has added development and opportunity dimensions with the HDI [13], but data quality varies by country [13]. In the future, a promising path would be an index that explicitly weights income distribution, education / health outcomes, and social mobility. Such an index would signal to policymakers that they should not only aim to raise incomes, but to also equal opportunities and access to resources. As well as this, graphical measures, such as the aforementioned Lorenz curve for other drivers, could also significantly increase our understanding of inequality within an economy, and allow for details that a single number would miss to be seen in a much more understandable way.

Notes

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The aim of the curve would be to provide a better picture of inequality, this isn't possible with current statistics, which don't report on all measures of inequality. Yet, if governments were to report on significantly more indicators, it would be possible to make such a curve reflect inequality to a much better degree, especially for such other factors, possibly even comparing between two factors, how an increase in 10 per cent in one factor affects another factor, inter-changing the x-axis and y-axis for other values. Currently it just demonstrates how life expectancy changes with income, yet income inequality, as stated previously, isn't a very good measure. ¹

References

- [1] OECD (n.d.) Income inequality. OECD Data and Statistics. <https://www.oecd.org/en/data/indicators/income-inequality.html>
- [2] Hasell, J. (2018) Measuring inequality: what is the Gini coefficient? Our World in Data. <https://ourworldindata.org/what-is-the-gini-coefficient>
- [3] International Monetary Fund (n.d.) Introduction to Inequality. IMF. <https://www.imf.org/en/Topics/Inequality/introduction-to-inequality>
- [4] United Nations Development Programme (UNDP) (n.d.) Human Development Index. Human Development Reports. <https://hdr.undp.org/data-center/human-development-index>
- [5] Eurostat (n.d.) Quality of life indicators: measuring quality of life. Statistics Explained. https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Quality_of_life_indicators_-_measuring_quality_of_life
- [6] Alfani, G. and Sardone, S. (2024) Long-term trends in income and wealth inequality in southern Italy: The Kingdom of Naples (Apulia), Sixteenth to Eighteenth Centuries. World Inequality Lab Working Paper, 2024/21. <https://wid.world/document/long-term-trends-in-income-and-wealth-inequality-in-southern-italy-world-inequality-lab>
- [7] Alfani, G. (2025) Income Inequality in History from the Preindustrial Medieval Era. Brewminate. <https://brewminate.com/income-inequality-in-history-from-the-preindustrial-medieval-era/>
- [8] World Population Review (n.d.) Gini Coefficient by Country 2025. WorldPopulationReview.com. <https://worldpopulationreview.com/country-rankings/gini-coefficient-by-country>
- [9] Palomino, J.C., Rodríguez, J.G. and Marrero, G.A. (2019) Intergenerational mobility in the US: One size doesn't fit all. CEPR (VoxEU column, 3 January 2019). <https://cepr.org/voxeu/columns/intergenerational-mobility-us-one-size-doesnt-fit-all>
<https://perma.cc/2UNG-F5LF>
- [10] Blesch, K., Hauser, O.P. and Jachimowicz, J.M. (2022) Measuring inequality beyond the Gini coefficient may clarify conflicting findings. *Nature Human Behaviour*, 6, 1525–1536. <https://doi.org/10.1038/s41562-022-01430-7>
- [11] Our World in Data (2025) Inequality-adjusted Human Development Index. <https://ourworldindata.org/grapher/inequality-adjusted-human-development-index>
- [12] United Nations Department of Economic and Social Affairs (n.d.) Sustainable Development Goal 10: Reduce inequality within and among countries. UN SDG. <https://sdgs.un.org/goals/goal10>
- [13] United Nations Development Programme (2023) Mitigating data challenges is imperative for human development and getting the SDGs back on track

- (Press Release, 26 April 2023). <https://www.undp.org/press-releases/mitigating-data-challenges-imperative-human-development-and-getting-sdgs-back-track>
- [14] Prieto, L.M., Flacke, J., Agüero-Valverde, J. and Van Maarseveen, M. (2018) Measuring Inequality of Opportunity in Access to Quality Basic Education: A Case Study in Florida, US. *ISPRS International Journal of Geo-Information*, 7(12), 465. <https://doi.org/10.3390/ijgi7120465>
 - [15] Social Progress Imperative (2025) Global Social Progress Index 2025. SocialProgress.org. <https://www.socialprogress.org/social-progress-index>
 - [16] Vosoughi, S., Roy, D. and Aral, S. (2018) The spread of true and false news online. *Science*, 359(6380): 1146–1151. <https://doi.org/10.1126/science.aap9559>
 - [17] Gao, J. et al. (2020) Mental health problems and social media exposure during COVID-19 outbreak. *PLOS ONE*, 15(4): e0231924. <https://doi.org/10.1371/journal.pone.0231924>
 - [18] World Health Organization (2020) Managing the COVID-19 infodemic: Promoting healthy behaviours and mitigating the harm from misinformation and disinformation (Joint statement, 23 Sep 2020). WHO. <https://www.who.int/news/item/23-09-2020-managing-the-covid-19-infodemic-promoting-healthy-behaviours-and-mitigating>
 - [19] Easterlin, R.A. (2010) The happiness–income paradox revisited. *Proceedings of the National Academy of Sciences*, 107(52): 22463–22468. <https://doi.org/10.1073/pnas.1015962107>
 - [20] Kahneman, D. and Deaton, A. (2010) High income improves evaluation of life but not emotional well-being. *Proceedings of the National Academy of Sciences*, 107(38): 16489–16493. <https://doi.org/10.1073/pnas.1011492107>
 - [21] Festinger, L. (1954) A theory of social comparison processes. *Human Relations*, 7(2): 117–140. [https://www.human-science.org/docs/Festinger%20\(1954\)%20A%20Theory%20of%20Social%20Comparison%20Processes.pdf](https://www.human-science.org/docs/Festinger%20(1954)%20A%20Theory%20of%20Social%20Comparison%20Processes.pdf)
 - [22] Ferrer-i-Carbonell, A. (2005) Income and well-being: An empirical analysis of the comparison income effect. *Journal of Public Economics*, 89(5–6): 997–1019. <https://doi.org/10.1016/j.jpubeco.2004.06.003>
 - [23] Petty, W. (1682) *Essays on Mankind and Political Arithmetic*. Project Gutenberg eBook #5619. <https://www.gutenberg.org/ebooks/5619>
 - [24] King, G. (1688) Social tables: Scheme for 1688 (early national social tables). (Digitized materials at NYU Abu Dhabi/University of York). <https://www.york.ac.uk/depts/maths/histstat/king.htm>
 - [25] Quetelet, A. (1835) *Sur l’homme et le développement de ses facultés, ou Essai de physique sociale*. Bruxelles: Bachelier.
 - [26] Lorenz, M.O. (1905) Methods of measuring the concentration of wealth. *Publ. of the ASA*, 9(70): 209–219. <https://www.jstor.org/stable/2965587>

- [27] Gini, C. (1912) Variabilità e mutabilità. Studi Economico-Giuridici della R. Università di Cagliari. (Ceriani 2012 commentary).
- [28] Cowell, F.A. (1980) Measuring Inequality. In Tufte, E.R. (ed.) *The Quantitative Analysis of Social Problems*. Reading, MA: Addison-Wesley. <https://piketty.pse.ens.fr/cowell1980.pdf>
- [29] Milanović, B. (2011) A short history of global inequality: The past two centuries. *Explorations in Economic History*, 47(3): 560–567. <https://doi.org/10.1016/j.eeh.2011.05.001>
- [30] Theil, H. (1967) *Economics and Information Theory*. North-Holland.
- [31] Cowell, F.A. (2003) Theil, inequality and the structure of income distribution. LSE Discussion Paper No. DARP 67. https://eprints.lse.ac.uk/2288/1/Theil,_Inequality_and_the_Structure_of_Income_Distribution.pdf
- [32] Vo, D.H. (2019) Information theory and an entropic approach to an analysis of fiscal inequality. *Entropy*, 21(7): 643. <https://doi.org/10.3390/e21070643>
- [33] Brunori, P., Ferreira, F.H.G. and Peragine, V. (2013) Inequality of Opportunity, Income Inequality and Economic Mobility: Some International Comparisons. IZA Discussion Paper No. 7155. <https://ftp.iza.org/dp7155.pdf>
- [34] Atkinson, A.B. (1970) On the measurement of inequality. *Journal of Economic Theory*, 2(3): 244–263.
- [35] Cowell, F.A. (2011) *Measuring Inequality*. 3rd edn. Oxford: Oxford University Press.
- [36] Shannon, C.E. (1948) A mathematical theory of communication. *Bell System Technical Journal*, 27(3,4): 379–423, 623–656.
- [37] Shorrocks, A.F. (1980) The class of additively decomposable inequality measures. *Econometrica*, 48(3): 613–625.
- [38] Sen, A. (1997) *On economic inequality*. Oxford: Clarendon Press (Expanded edition).
- [39] Gastwirth, J.L. (1972) The estimation of the Lorenz curve and Gini index. *Review of Economics and Statistics*, 54(3): 306–316.
- [40] Deininger, K. and Squire, L. (1996) A new data set measuring income inequality. *World Bank Economic Review*, 10(3): 565–591.
- [41] World Bank (n.d.) World Development Indicators: Gini Index (World Bank estimate). World Bank. <https://databank.worldbank.org/source/world-development-indicators>
- [42] Corak, M. (2012) Inequality from generation to generation: The United States in comparison. In R.R. Rycroft (ed.) *The Economics of Inequality, Poverty and Discrimination in the 21st Century*. Santa Barbara, CA: ABC-CLIO.
- [43] CORE Project (2017) *The Economy*. Cambridge, MA: MIT Press (Open Access).

- [44] Office for National Statistics (2020) Household income inequality, UK: Financial year ending 2020 (provisional). ONS. <https://www.ons.gov.uk/peoplepopulationandcommunity/personalandhouseholdfinances/incomeandwealth/bulletins/householdincomeinequalityfinancial/financialyearending2020provisional>
- [45] Lambert, P.J. (2001) The distribution and redistribution of income. 3rd edn. Manchester: Manchester University Press.
- [46] Sen, A. (1999) Development as Freedom. Oxford: Oxford University Press.
- [47] Stiglitz, J.E., Sen, A. and Fitoussi, J.-P. (2009) Report by the Commission on the Measurement of Economic Performance and Social Progress. Paris: Commission. https://ec.europa.eu/environment/beyond_gdp/stiglitz_sen_fitoussi_report.pdf
- [48] UNDP (2010) Human Development Report 2010: The Real Wealth of Nations – Pathways to Human Development. New York: Palgrave Macmillan.
- [49] Ferreira, F.H.G. and Gignoux, J. (2011) The measurement of inequality of opportunity: Theory and an application to Latin America. *Review of Income and Wealth*, 57(4): 622–657.
- [50] Klugman, J., Rodríguez, F. and Choi, H.-J. (2011) The HDI 2010: New controversies, old critiques. *Journal of Economic Inequality*, 9: 249–288.
- [51] United Nations (2015) Transforming Our World: The 2030 Agenda for Sustainable Development. New York: United Nations (A/RES/70/1). <https://undocs.org/en/A/RES/70/1>
- [52] Chetty, R., Hendren, N., Kline, P. and Saez, E. (2014) Where is the Land of Opportunity? The Geography of Intergenerational Mobility in the United States. *Quarterly Journal of Economics*, 129(4): 1553–1623.
- [53] Jerven, M. (2013) Poor Numbers: How We Are Misled by African Development Statistics and What to Do About It. Ithaca: Cornell University Press.
- [54] Foster, J., López-Calva, L.F. and Székely, M. (2009) Measuring the distribution of well-being and poverty. *Journal of Economic Inequality*, 7: 1–26.
- [55] Ura, K., Alkire, S., Zangmo, T. and Wangdi, K. (2012) A Short Guide to the Gross National Happiness Index. Thimphu: Centre for Bhutan Studies.
- [56] Easterly, W. (2001) The Elusive Quest for Growth: Economists’ Adventures and Misadventures in the Tropics. Cambridge, MA: MIT Press.
- [57] Kahneman, D., Diener, E. and Schwarz, N. (1999) Well-Being: The Foundations of Hedonic Psychology. New York: Russell Sage Foundation.
- [58] Lazer, D.M.J. et al. (2018) The science of fake news. *Science*, 359(6380): 1094–1096.
- [59] Financial Times (n.d.) The art of persuasion: how top AI chatbots can change your mind. FT.com. <https://www.ft.com/content/31e528b3-9800-4743-af0a-f5c3b80032d0>
- [60] Wikipedia “Easterlin paradox”. https://en.wikipedia.org/wiki/Easterlin_paradox

- [61] Easterlin, R.A. (1974) Does economic growth improve the human lot? Some empirical evidence. In P.A. David and M.W. Reder (eds.) Nations and Households in Economic Growth. New York: Academic Press, pp. 89–125.
- [62] Easterlin, R.A. (1995) Will raising the incomes of all increase the happiness of all? Journal of Economic Behavior & Organization, 27(1): 35–47.
- [63] Office for National Statistics (n.d.) Healthy life expectancy by national area deprivation, England and Wales: 2013 to 2015 and 2020 to 2022. ONS. <https://www.ons.gov.uk/peoplepopulationandcommunity/healthandsocialcare/healthinequalities/bulletins/healthylifeexpectancybynationalareadeprivationenglandandwales/between2013to2015and2020to2022>
- [64] Thinley, J. (2010) Gross National Happiness and Development. Thimphu: Centre for Bhutan Studies.
- [65] Wikipedia “Richard Easterlin”. https://en.wikipedia.org/wiki/Richard_Easterlin
- [66] OECD (2017) How’s Life? 2017: Measuring Well-being. Paris: OECD Publishing. https://doi.org/10.1787/how_life-2017-en
- [67] Office for National Statistics (2025) Household income inequality, UK: financial year ending 2024. Statistical bulletin. <https://www.ons.gov.uk/peoplepopulationandcommunity/personalandhouseholdfinances/incomeandwealth/bulletins/householdincomeinequalityfinancial/financialyearending2024>
- [68] MacKay, D. J. C. (2003) *Information Theory, Inference, and Learning Algorithms*. Cambridge: Cambridge University Press.